COGS 9 Section

TA: Lindsey Gu
Fri October 30

Attendance!

webclicker.web.app



Code: OMFZUC

How did you feel about this week?

A. 😄
B. 🙂
C. 😬
D. 😭

Reminders

- Group project: Checkpoint 1 was due this Wednesday!
  - If you haven't turned it in, please do so ASAP to avoid more late penalty
- Assignment 1 on **Data Visualization** is due **Wed Week7**
  - Find assignment document on website
  - Due on Gradescope individually
  - Please start early!
- R4 quiz due **next Wednesday**

Questions?

Core Ideas

"Data is Personal"

Data is Personal: Attitudes and Perceptions of Data Visualization in Rural Pennsylvania

Evan M. Peck
Bucknell University
evan.peck@bucknell.edu

Sofia E. Ayuso
Bucknell University
sea018@bucknell.edu

Omar El-Etr
Bucknell University
omar.eletr@bucknell.edu

ABSTRACT

Many of the guidelines that inform how designers create data visualizations originate in studies that unintentionally exclude populations that are most likely to be among the "data poor". In this paper, we explore which factors may drive attention and trust in rural populations with diverse economic and educational backgrounds - a segment that is largely underrepresented in the data visualization literature. In 42 semi-structured interviews in rural Pennsylvania (USA), we find that a complex set of factors intermix to inform attitudes and perceptions about data visualization - including educational background, political affiliation, and personal experience. The data and materials for this research can be found at https://osf.io/uxwts/

CCS CONCEPTS

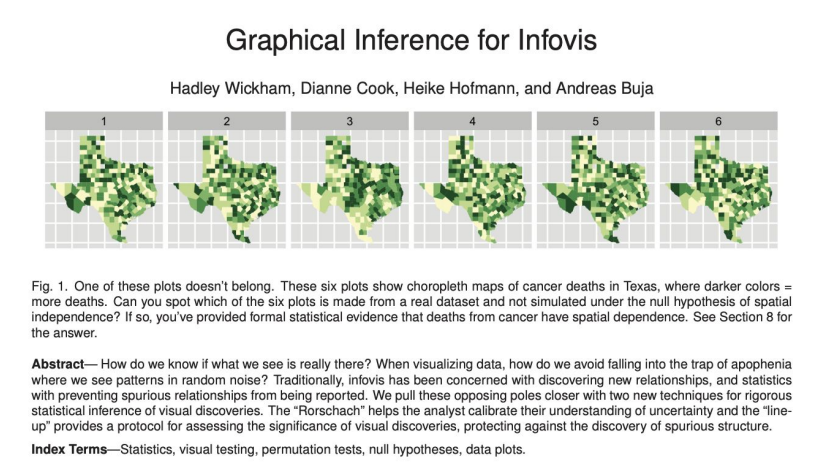
•**Human-centered computing** → *Visualization theory, concepts and paradigms*;



- **Data Visualization is not neutral.** People interpret charts through various factors
- **Trust in sources varies,** and it can align with education and political identity
- **First impressions** are extremely important because in this case study people didn't change their opinion after the sources were revealed, they based their opinion only on the visual data

Core Ideas

“Graphical Inference for Infovis”



Graphical Inference for Infovis

Hadley Wickham, Dianne Cook, Heike Hofmann, and Andreas Buja

Fig. 1. One of these plots doesn’t belong. These six plots show choropleth maps of cancer deaths in Texas, where darker colors = more deaths. Can you spot which of the six plots is made from a real dataset and not simulated under the null hypothesis of spatial independence? If so, you’ve provided formal statistical evidence that deaths from cancer have spatial dependence. See Section 8 for the answer.

Abstract— How do we know if what we see is really there? When visualizing data, how do we avoid falling into the trap of apophenia where we see patterns in random noise? Traditionally, infovis has been concerned with discovering new relationships, and statistics with preventing spurious relationships from being reported. We pull these opposing poles closer with two new techniques for rigorous statistical inference of visual discoveries. The “Rorschach” helps the analyst calibrate their understanding of uncertainty and the “line-up” provides a protocol for assessing the significance of visual discoveries, protecting against the discovery of spurious structure.

Index Terms—Statistics, visual testing, permutation tests, null hypotheses, data plots.

- **Apophenia** is a core risk in visualization. Without skepticism you can “discover” patterns that won’t replicate
- Line-ups operationalize visual significance by hiding the real plot among null plots.
- Null hypotheses depend on the plot’s question. **Choosing the right null** is a central design decision for graphical inference.

Data Visualization

Visual encodings for quantitative data

Quantitative encodings show the *magnitude* of a value

From **most to least** precise:

1. **Position:** value is shown by where a mark is placed along a scale
2. **Length:** value is shown by how long or tall a mark is
3. **Angle:** value is shown by size of an angle
4. **Area:** value is shown by how much 2D space a mark covers
5. **Color:** value is shown by differences in saturation or hue

Visual encodings for qualitative data

Qualitative encodings distinguish *categories*

1. **Position:** separate categories by spatial placement or grouping
2. **Color:** different colors for different categories
3. **Shape:** different shapes for different categories
4. **Texture/pattern:** stripes, dots, etc., to distinguish categories

Think like a Data Analyst

Use this framework to evaluate any chart

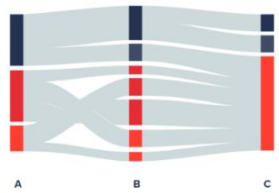
Ask yourself:

- What is the main message?
- Can I compare values easily?
- Is anything misleading?
- Is this the right chart type?

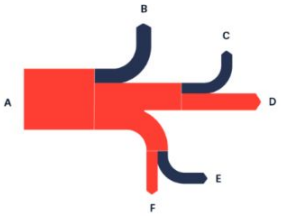
Recall what was taught in lectures!

Good Visualizations

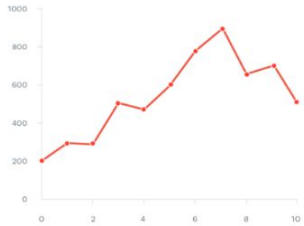
Alluvial Diagram



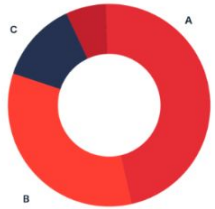
Sankey Diagram



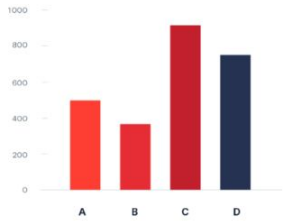
Line Graph



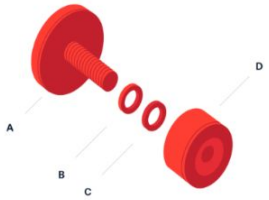
Donut Chart



Bar Chart (Vertical)



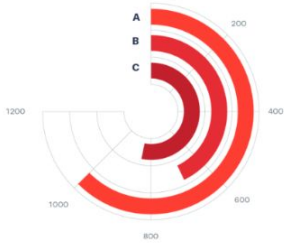
Exploded View Drawing



Polar Area Chart



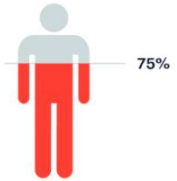
Radial Bar Chart



Sunburst Diagram



Pictorial Percentage Chart

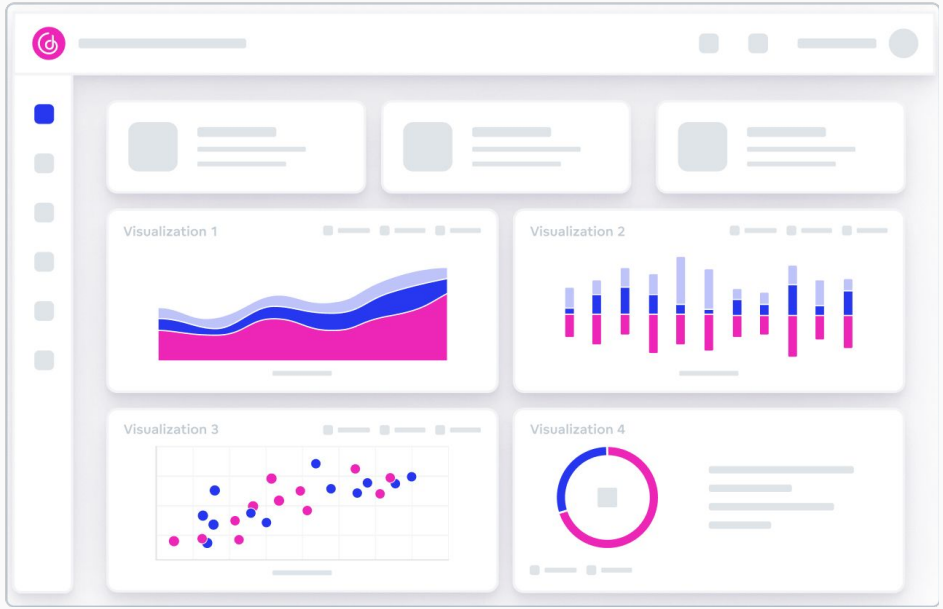


Learn more here: https://datavizproject.com/

Interactive Data Visualization

Interactive data visualization lets users explore data instead of only viewing one fixed chart.

- Static charts show one fixed view of the data
- Interactive charts let users hover, click, filter, zoom, and explore
- Users can answer their own questions instead of only following one path



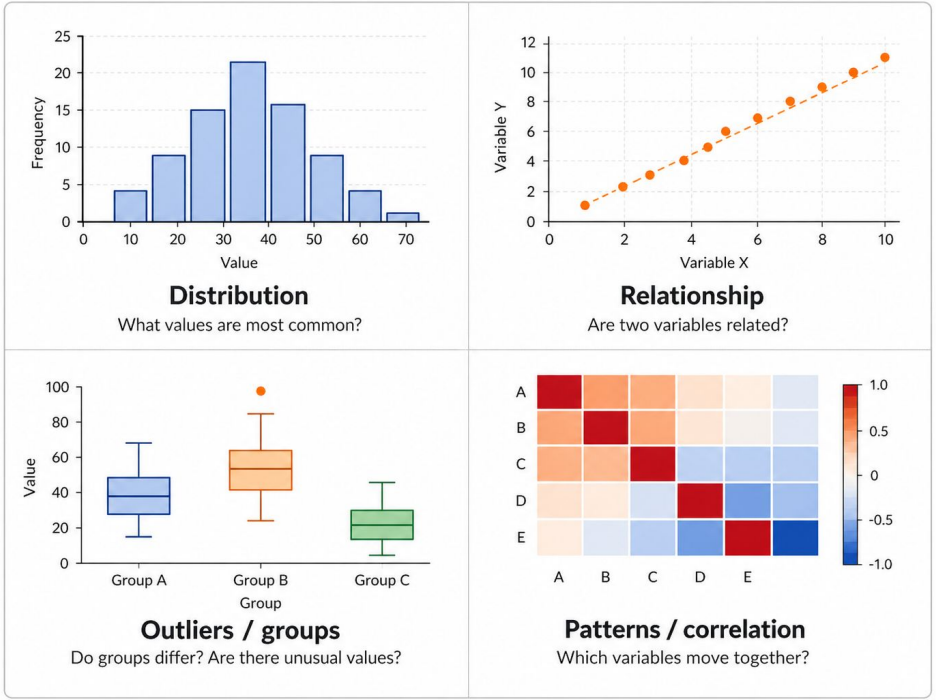
EDA: Finding Patterns Before Making Conclusions

EDA uses visualizations to understand a dataset before making claims or building models.

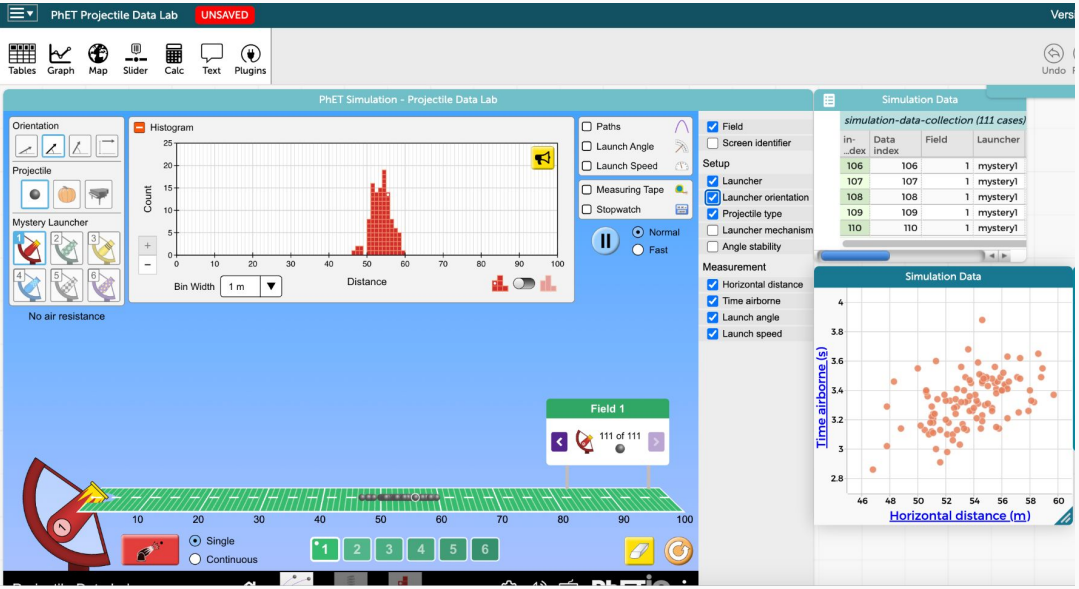
Common EDA visuals: histograms, scatterplots, box plots, and heatmaps

EDA helps us ask:

- What patterns do we notice?
- Are there outliers?
- Are groups different?
- Are two variables related?
- Is anything missing or unusual?



Interactive Activity: PhET Projectile Data Lab



Open Document or Browse Example → PhET Projectile Data Lab → Open → Variability → Start playing with it

- Lets users change variables and instantly see the data change
- Supports EDA through tables, graphs, and experimentation

https://codap.concord.org/app/static/dg/en/cert/

Data Visualization Checklist

by Stephanie Evergreen & Ann K. Emery
May 2016

This checklist is meant to be used as a guide for the development of high impact data visualizations. Rate each aspect of the data visualization by circling the most appropriate number, where 2 points means the guideline was fully met, 1 means it was partially met, and 0 means it was not met at all. n/a should not be used frequently, but reserved for when the guideline truly does not apply. For example, a pie chart has no axes lines or tick marks to rate. If the guidelines has been broken intentionally to make a point, rate it n/a and deduct those points from the total possible. Refer to the Data Visualization Anatomy Chart on the last page for guidance on vocabulary and the Resources at the end for more details.

https://stephanieevergreen.com/wp-content/uploads/2016/10/DataVizChecklist_May2016.pdf

The New York Times

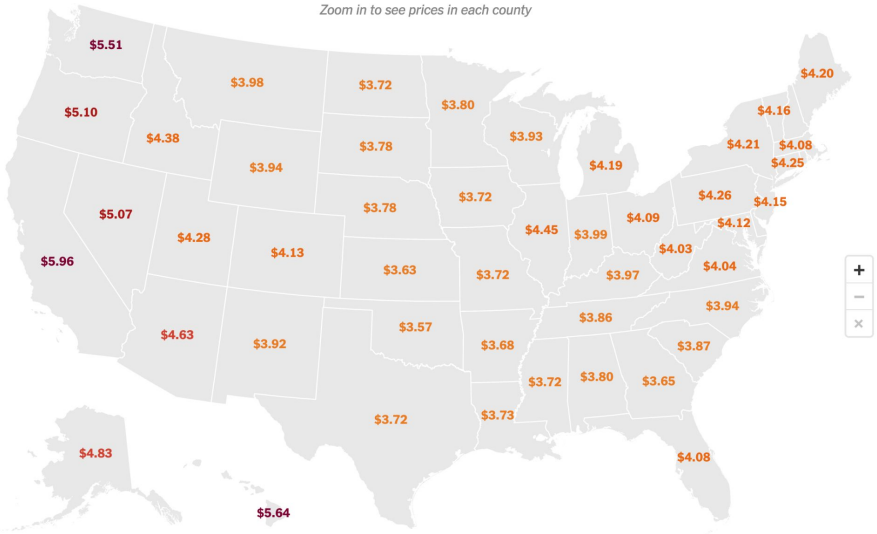
How High Are Gas Prices Where You Live?

By Matthew Bloch, Steven Rich and Emmett Lindner

Updated April 28, 2026 at 6:58 a.m. E.T.

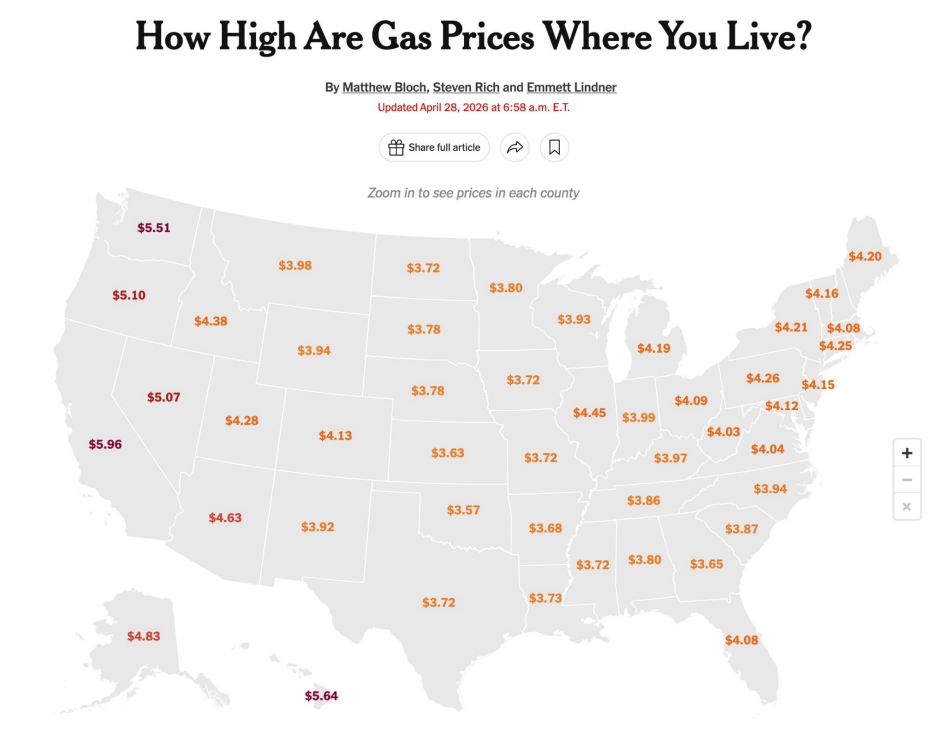
Share full article

Zoom in to see prices in each county





The New York Times

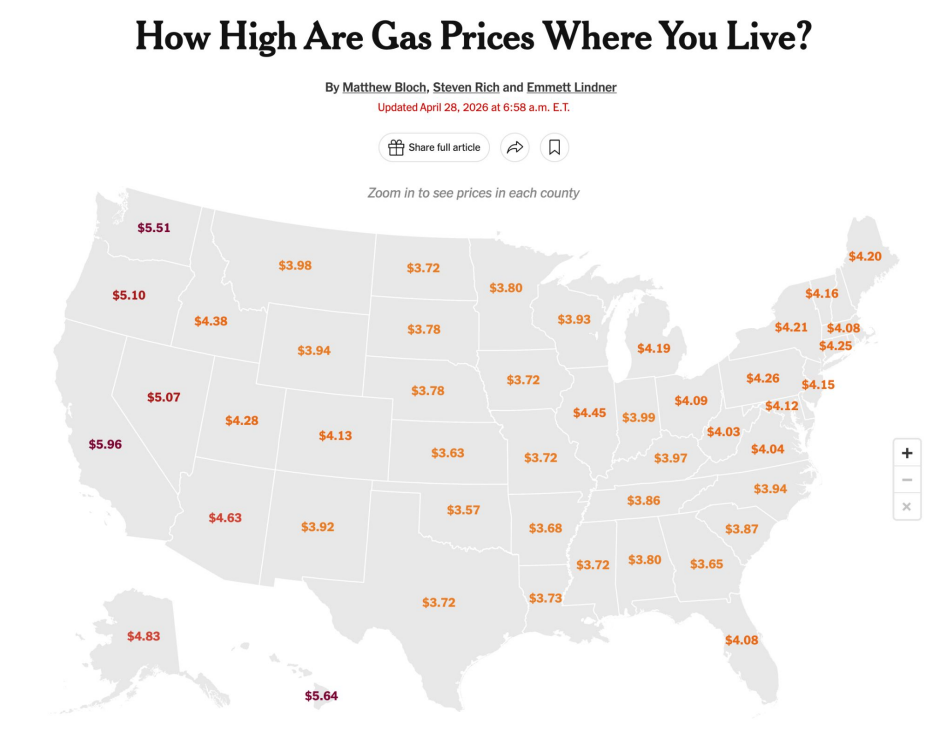


Data Type

Quantitative (Gas Prices per Gallon in US Dollars)

What visual encodings are used?

The New York Times



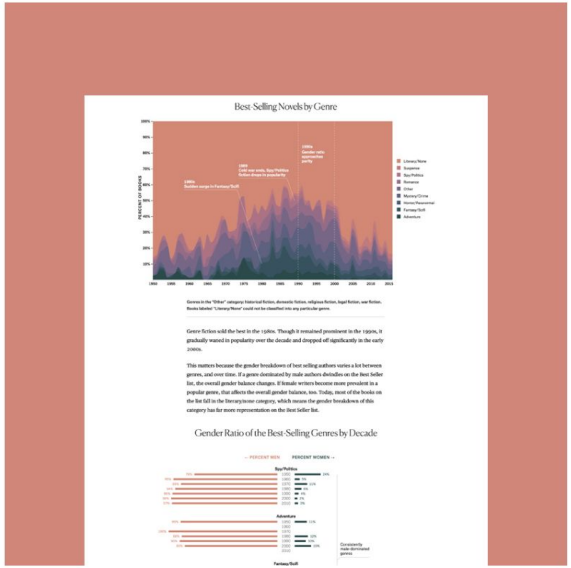
Visual Encoding

Position: Each price is listed spatially where that state is

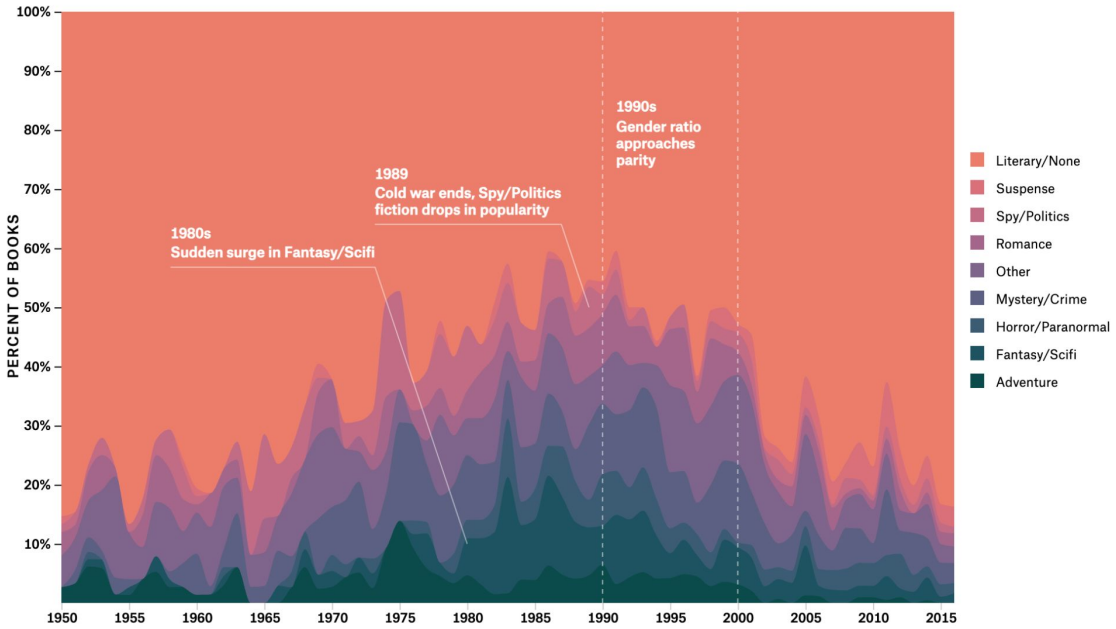
Color: Red being the highest prices and orange for mid-range

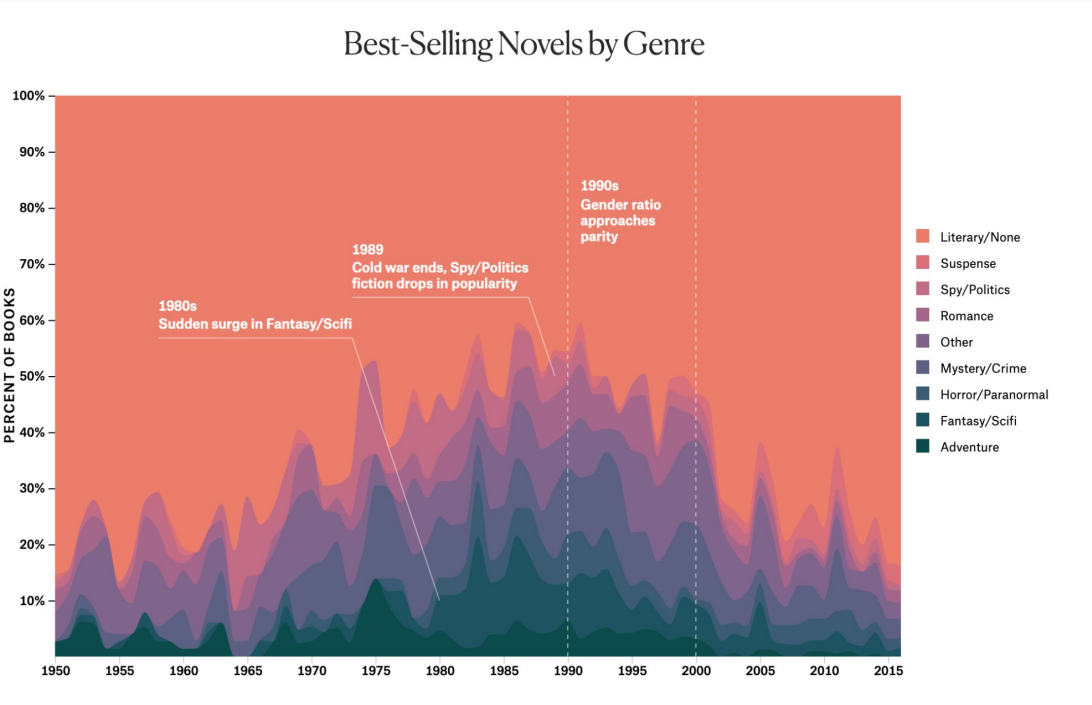
#25

JUN 2017



Best-Selling Novels by Genre



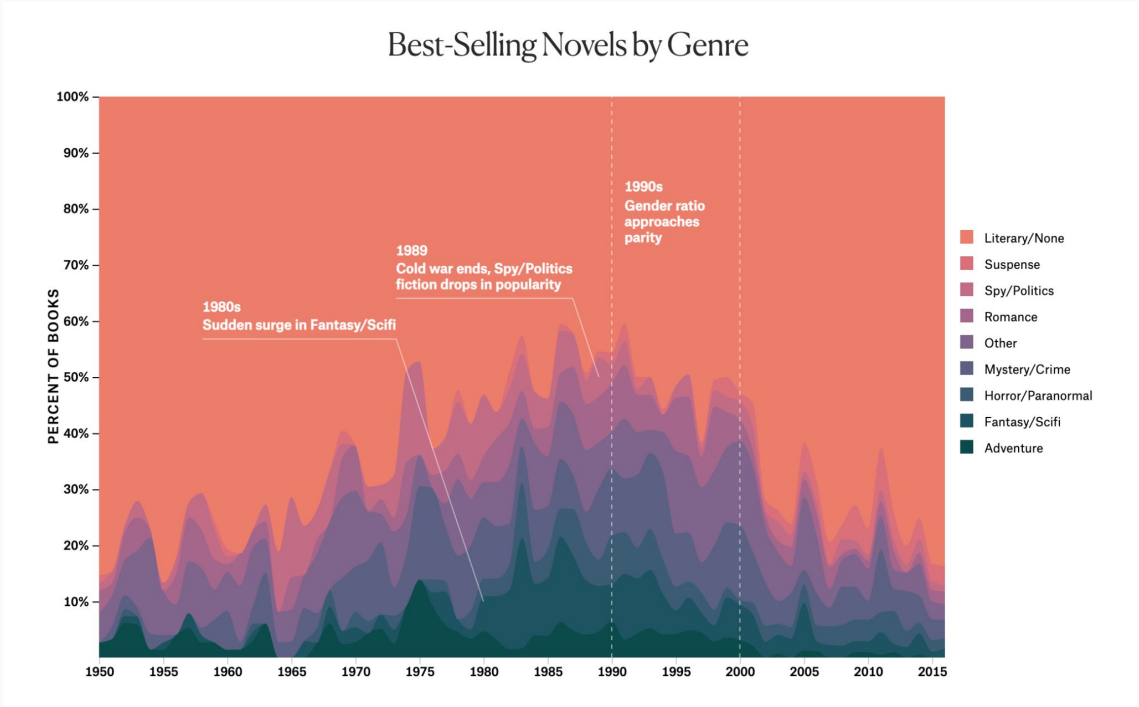


Data Types

Quantitative (% of best-selling novels)

Categorical (Genre over time)

What visual encodings are used?



Visual Encodings

Position: Y-axis encodes share of books, X-axis encodes time (1950 to 2015)

Color: Each hue in the legend corresponds to a distinct genre

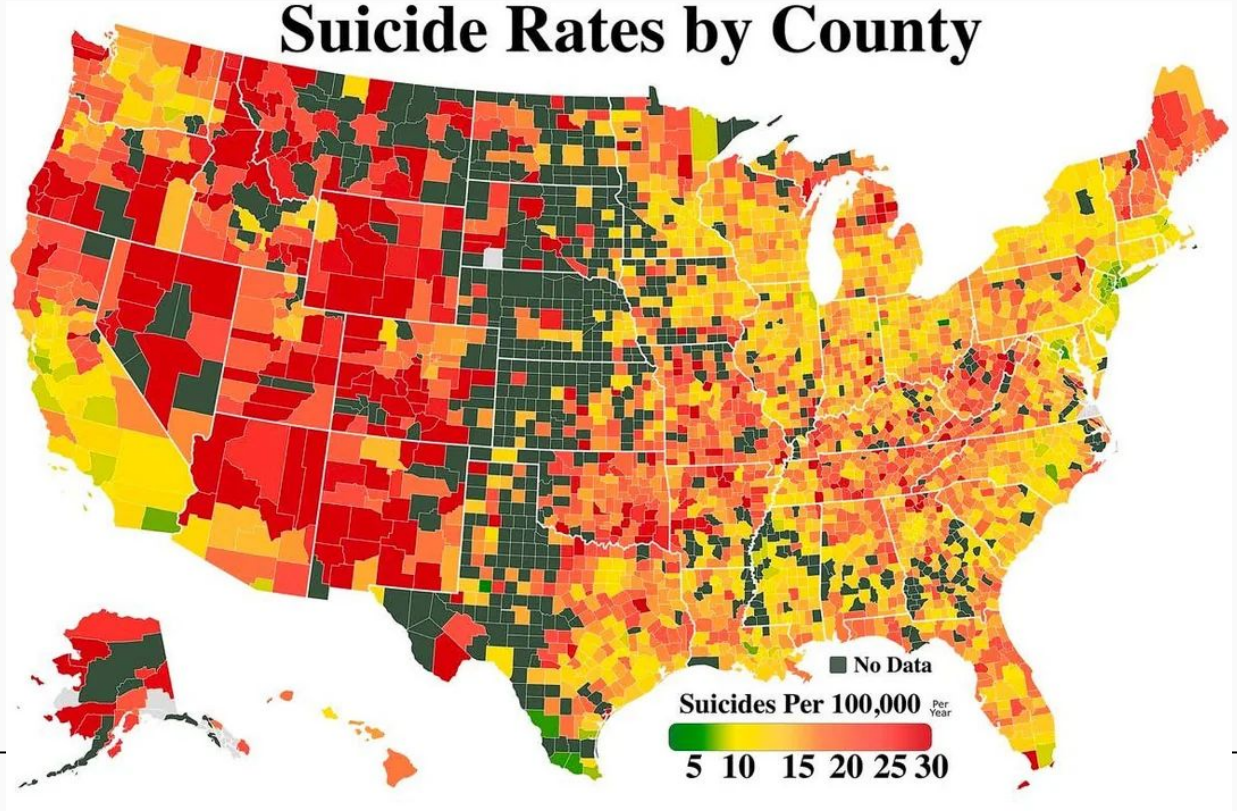
Length/Area: The thickness of each stacked band shows that genre's share of best-sellers.

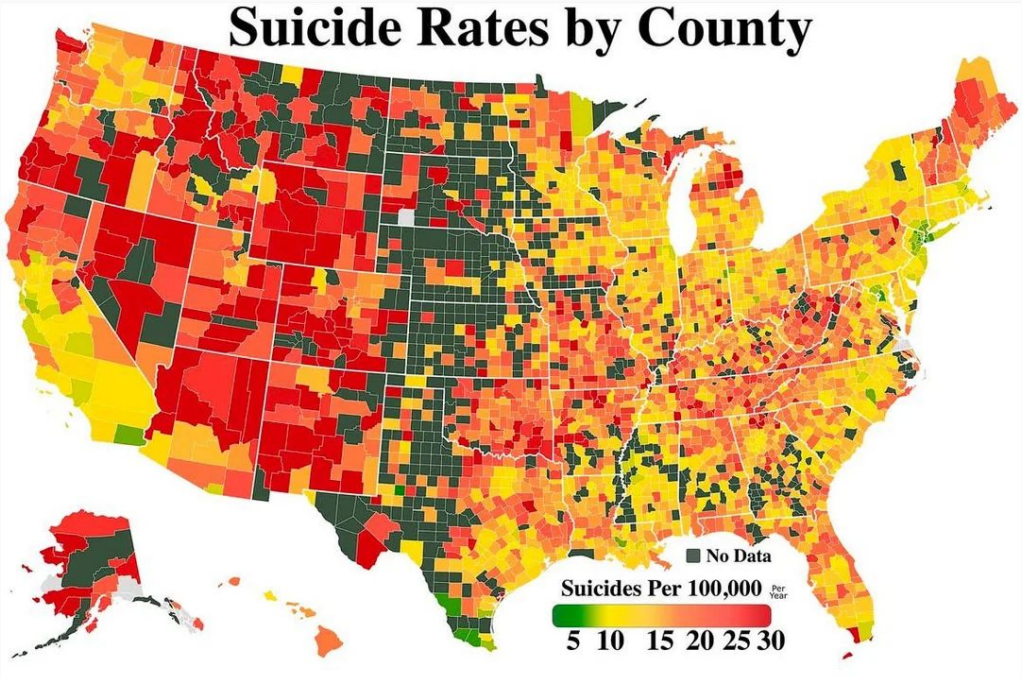
Annotation: Callouts and dashed vertical lines mark significant moments.

DATA VISUALIZATION

The Bad

What's Wrong With This?

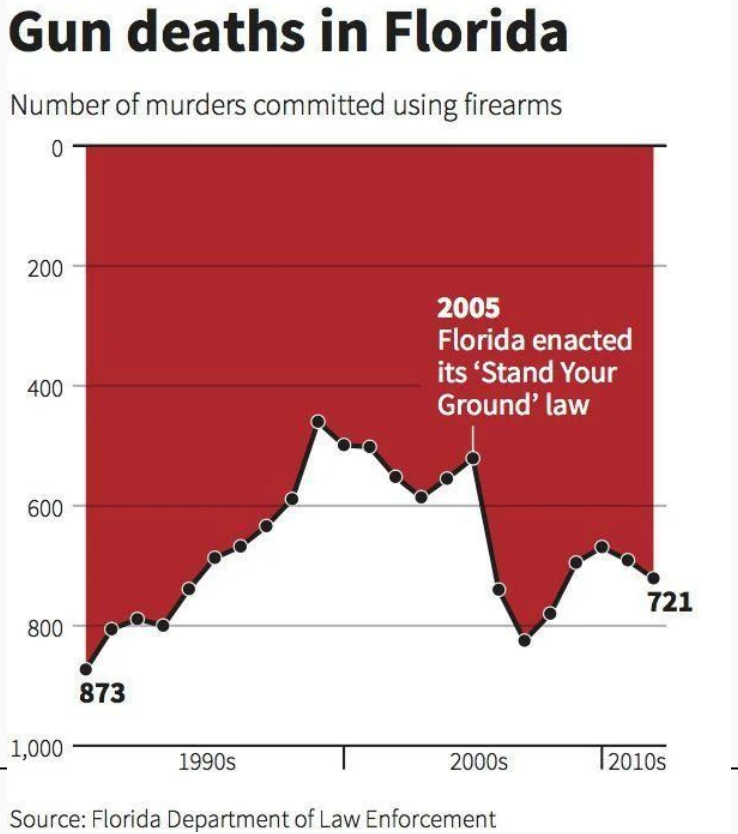


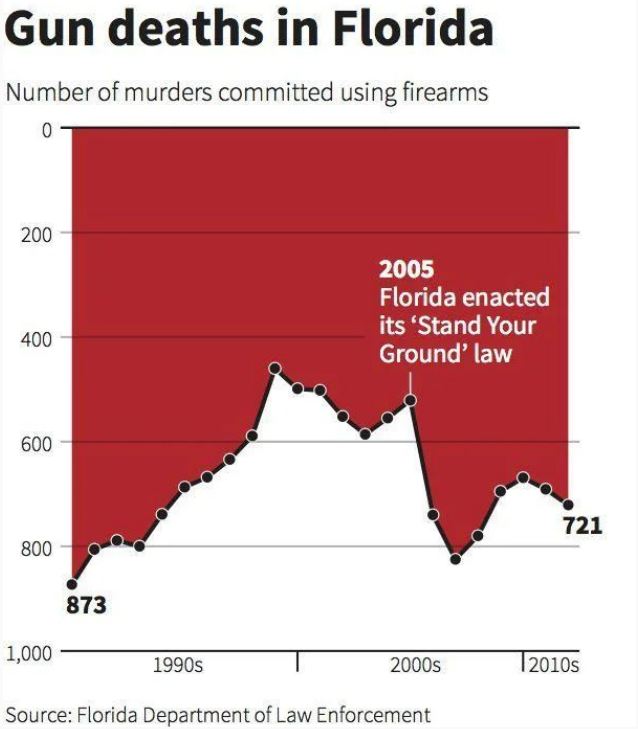


What's Wrong With This?

The greyish green used for "No Data" counties blends in with the low-rate end of the color scale, making it easy to misread missing data as simply having low suicide rates.

What's Wrong With This?

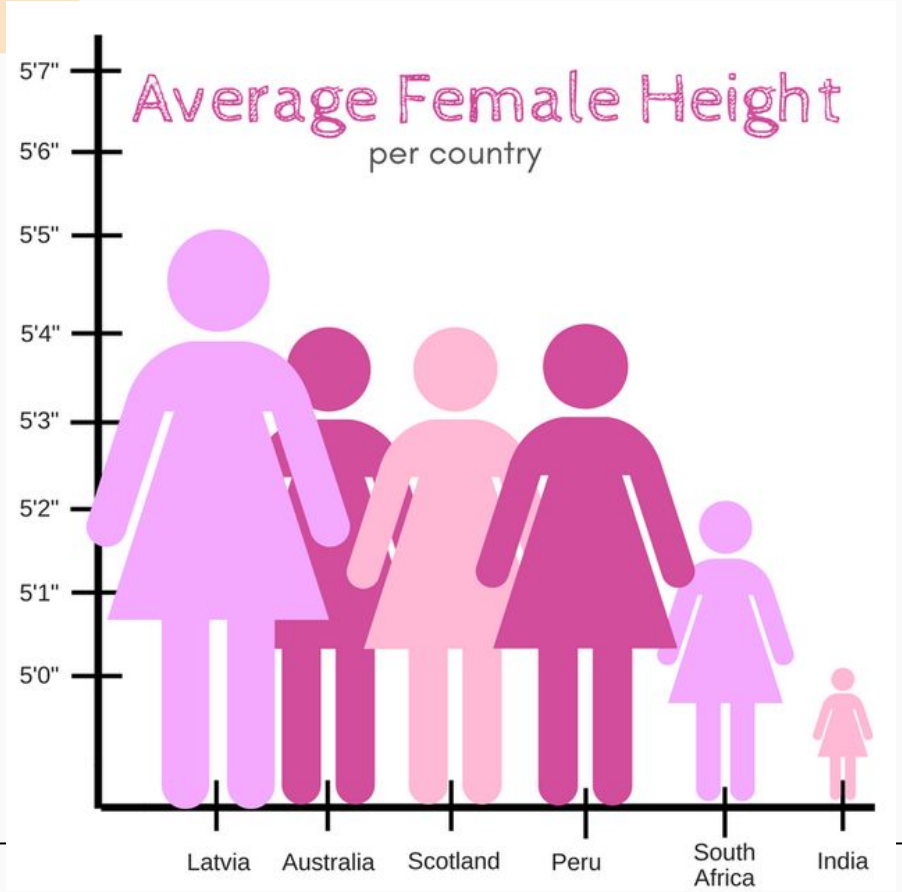


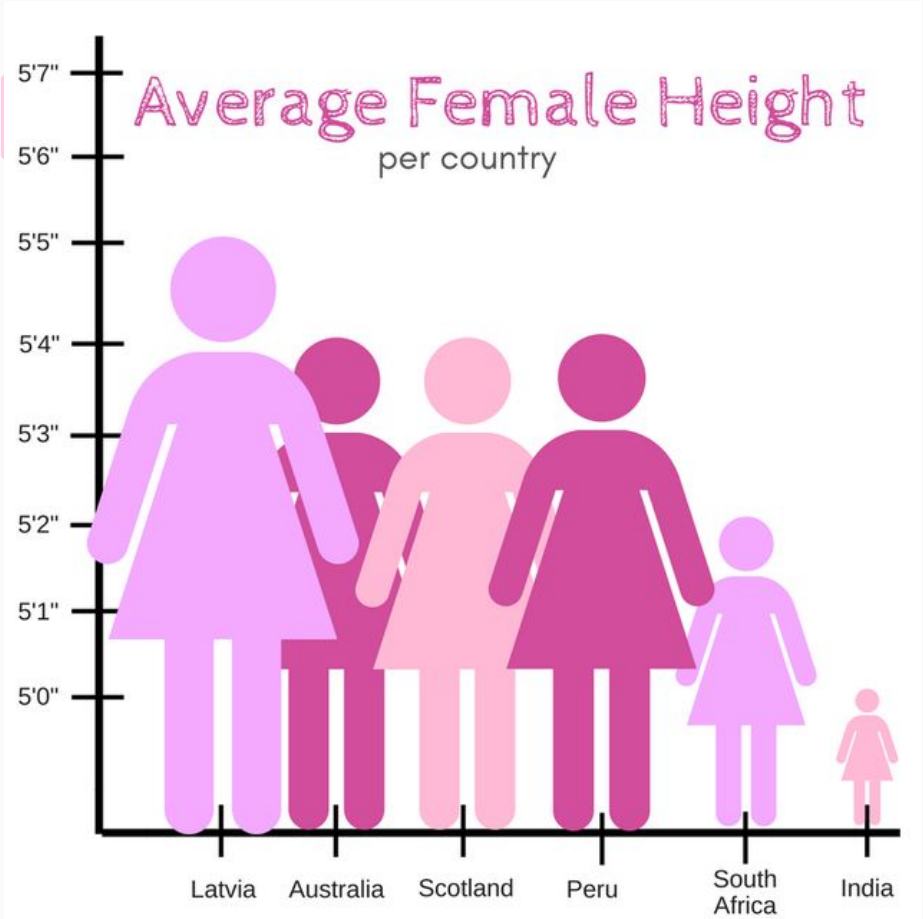


What's Wrong With This?

This graph is misleading. It inverts the y-axis making it appear as though Gun death's in Florida decreased after the "Stand Your Ground" law was enacted (even though the opposite happened in actuality)

What's Wrong With This?



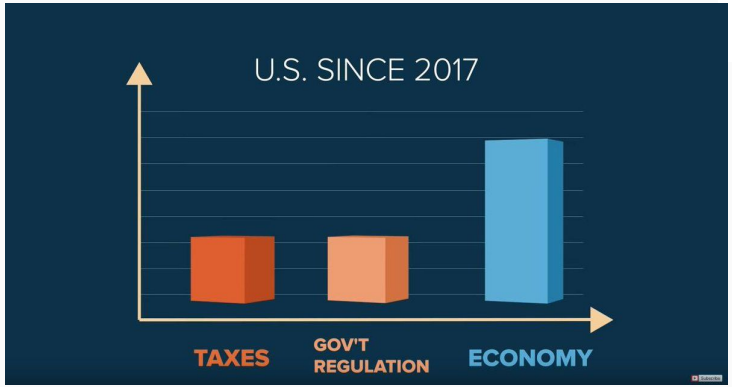


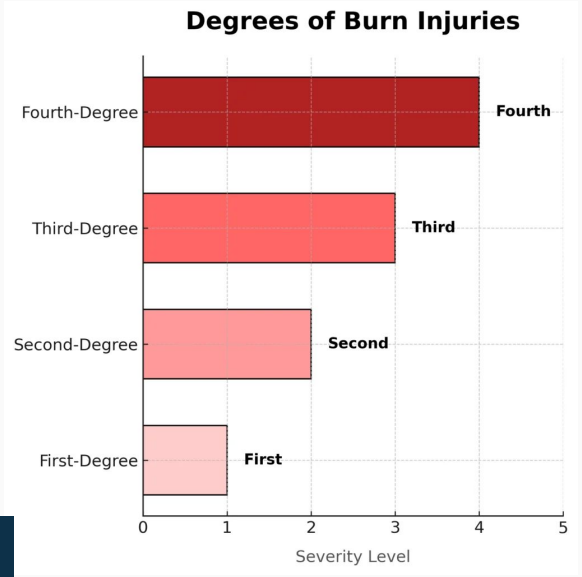
What's Wrong With This?

The truncated y-axis exaggerates differences, and scaling pictograms by height also inflates their width, so the perceived size gap grows much faster than the actual data difference.

The REALLY Bad





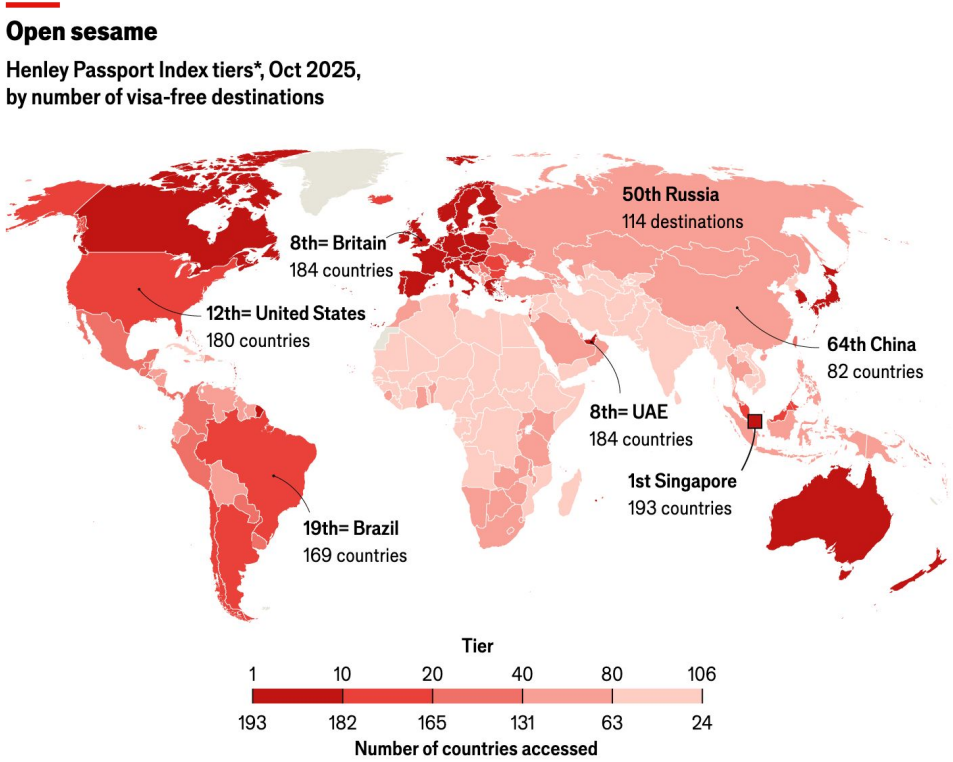


Source: www.reddit.com/r/dataisugly/

Data Journalism

“Data journalism or data-driven journalism (DDJ) is journalism based on the filtering and analysis of large data sets for the purpose of creating or elevating a news story.” (wikipedia)

Data Journalism



https://www.economist.com/graphic-detail/2025/10/16/the-worlds-most-and-least-powerful-passports

Data Journalism



https://www.theguardian.com/food/ng-interactive/2022/apr/14/climate-crisis-food-systems-not-ready-biodiversity

Research about how to tell a data story

The design space for a data story

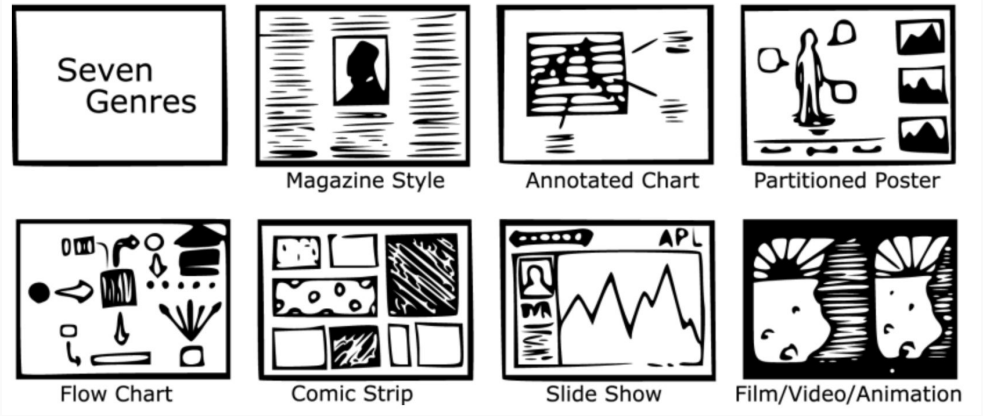


Table 1. Properties of Author-Driven and Reader-Driven Stories. Most visualizations lie along a spectrum between these two extremes.

Author-Driven	Reader-Driven
Linear ordering of scenes	No prescribed ordering
Heavy messaging	No messaging
No interactivity	Free interactivity

Segel, Edward, and Jeffrey Heer. "Narrative visualization: Telling stories with data." IEEE transactions on visualization and computer graphics 16, no. 6 (2010): 1139-1148.

Research about how to tell a data story

How to use animation affects data stories



http://kineticharts.idvxlab.com/#

Lan, Xingyu, Yang Shi, Yanqiu Wu, Xiaohan Jiao, and Nan Cao. "Kineticharts: Augmenting affective expressiveness of charts in data stories with animation design." *IEEE Transactions on Visualization and Computer Graphics* 28, no. 1 (2021): 933-943.

Personal Data

Besides data about the world, there are also data about ourselves, from health metrics and activity logs to the records of the doors you open throughout the day.

Personal Data

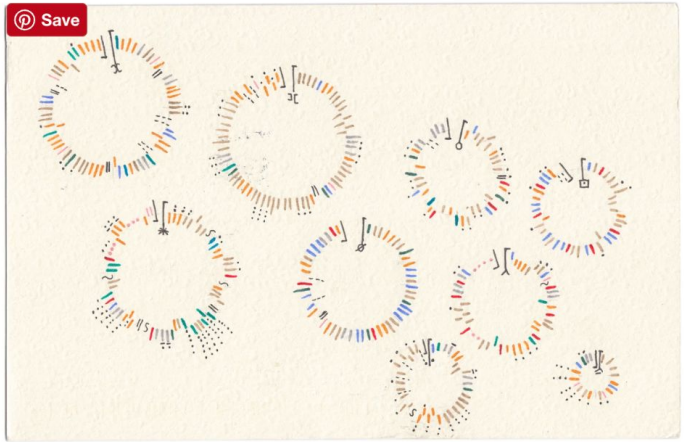
Example: **Dear Data**

a year-long project where the two information designers sent each other hand-drawn data postcards weekly, mapping their daily lives.

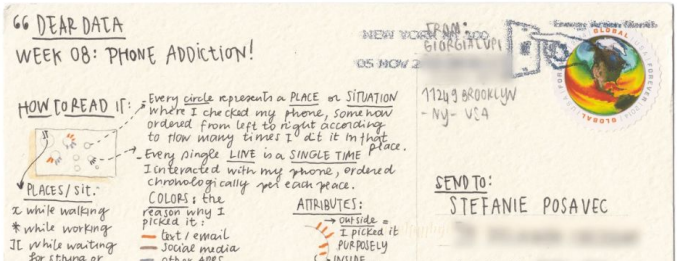
https://www.dear-data.com/all

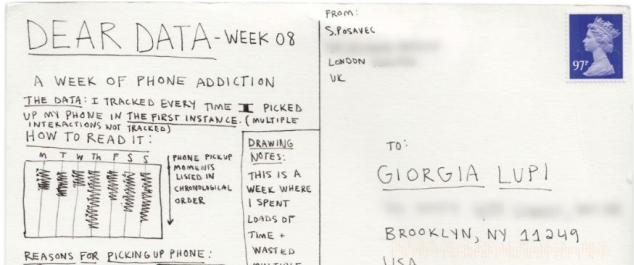
Personal Data

Week 08:
A week of phone addiction





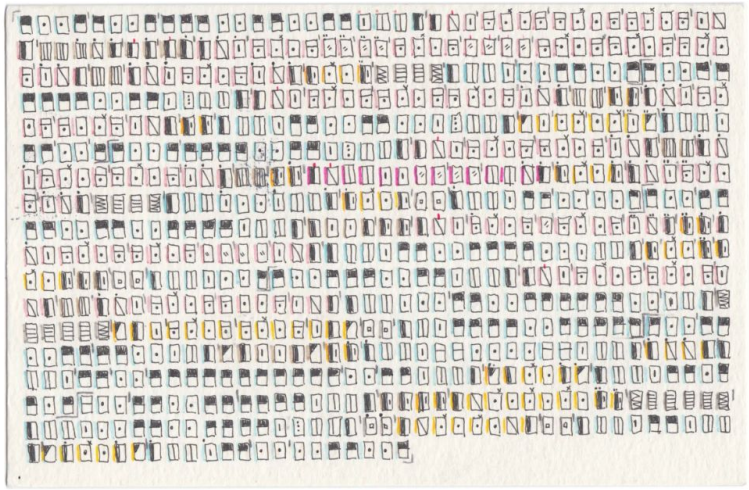


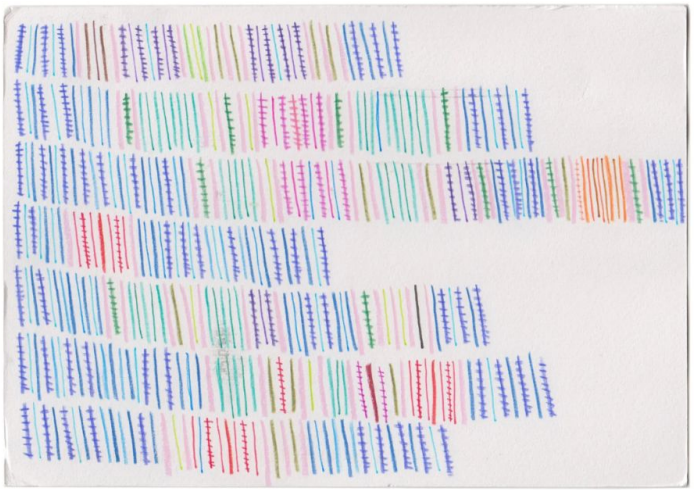


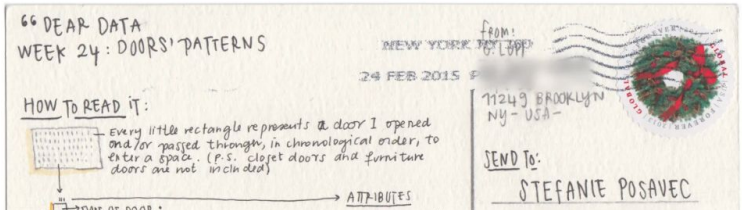
https://www.dear-data.com/week-24-a-week-of-doors

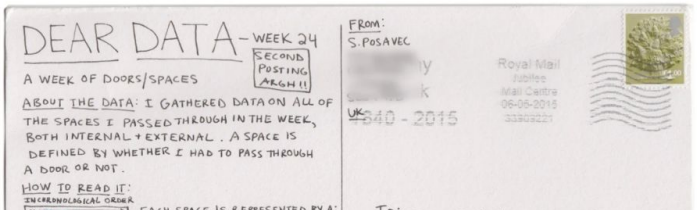
Personal Data

Week 24:
A week of doors/spaces

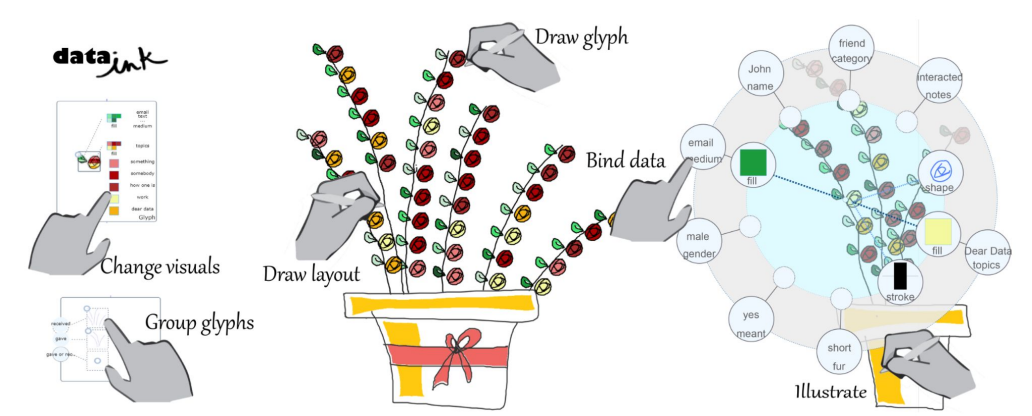


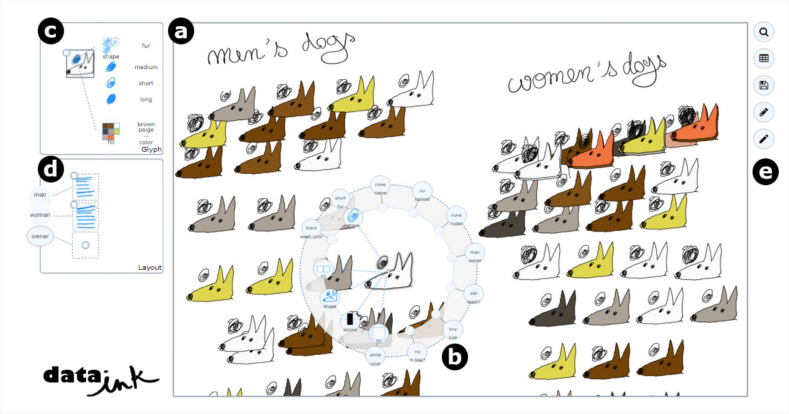






Research about authoring data vis/art





Xia, Haijun, Nathalie Henry Riche, Fanny Chevalier, Bruno De Araujo, and Daniel Wigdor. "DataInk: Direct and creative data-oriented drawing." In Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, pp. 1-13. 2018.

Data Physicalization

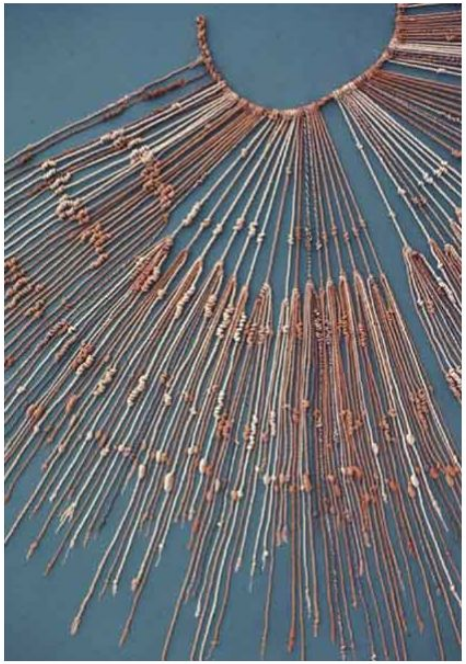
Data is beyond pixels!

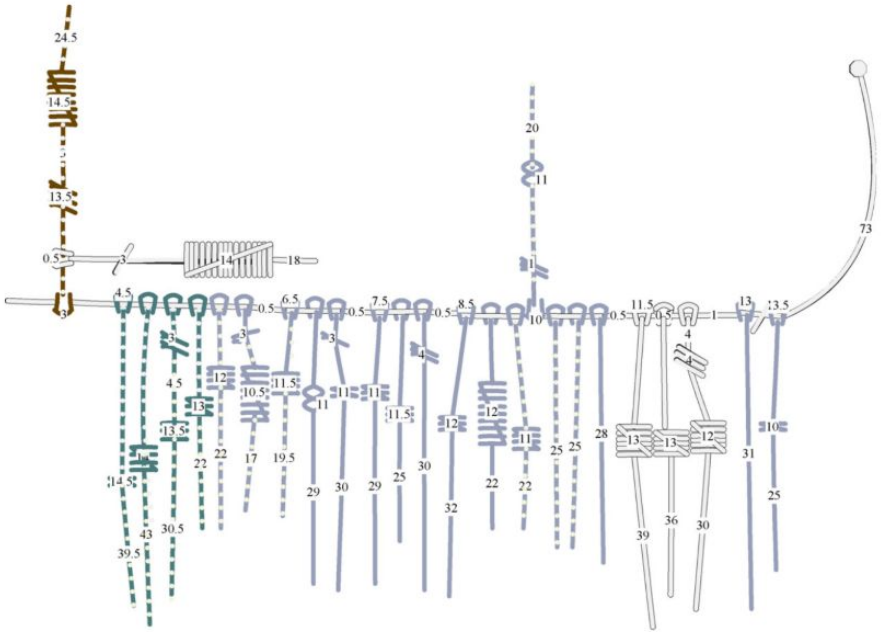
Data physicalization aims to help people explore, understand, and communicate data using computer-supported physical data representations.

https://dataphys.org/

Data Physicalization

2600 BC – Quipus



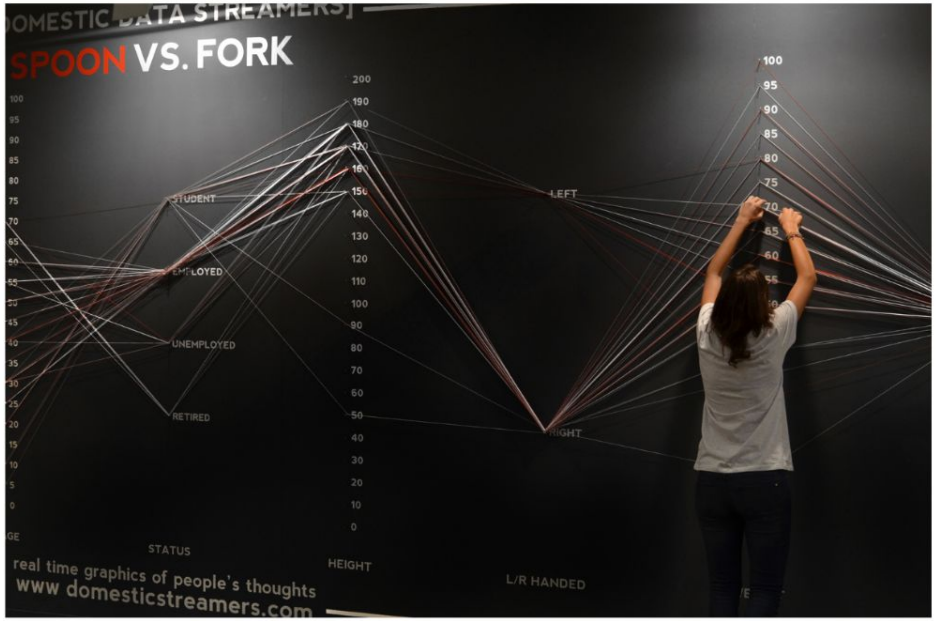


https://dataphys.org/list/peruvian-quipus/

Data Physicalization

2014 – Data Strings: Physical Parallel Coordinates





https://dataphys.org/list/data-strings-physical-parallel-coordinates/

Data Physicalization

2019 – Sleep Blanket



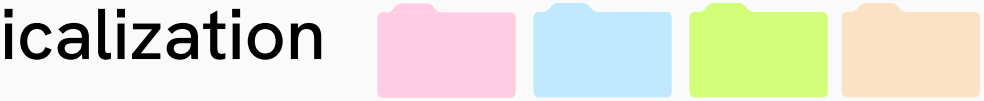




> *A visualization of my son's sleep pattern from birth to his first birthday. Crochet border surrounding a double knit body. Each row represents a single day. Each stitch represents 6 minutes of time spent awake or asleep.*

https://dataphys.org/list/sleep-blanket/

Research about data physicalization



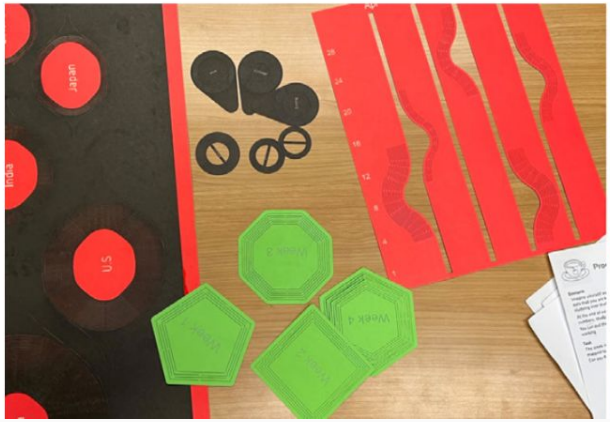
One of the advantages of data physicalization is allowing people to touch and feel data, and construct charts with tangible objects.

- Installation art
- Data / visualization literacy (help learning!)
- Informing design opportunities for digital tools

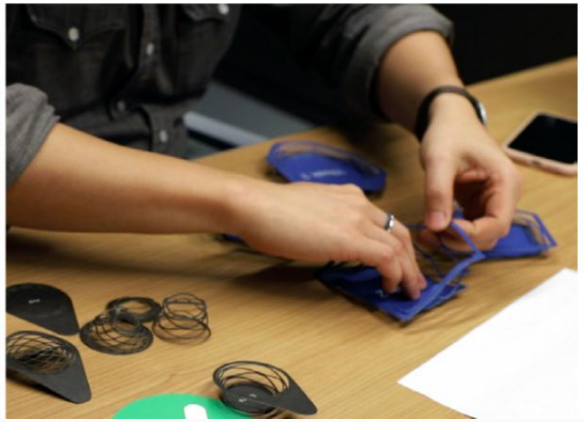
Research about data physicalization

KiriPhys: Exploring New Data Physicalization Opportunities

Foroozan Daneshzand, Charles Perin, Sheelagh Carpendale







Daneshzand, Foroozan, Charles Perin, and Sheelagh Carpendale. "Kiriphys: Exploring new data physicalization opportunities." IEEE Transactions on Visualization and Computer Graphics 29, no. 1 (2022): 225-235.

Research about data physicalization

Constructing Visual Representations: Investigating the Use of Tangible Tokens

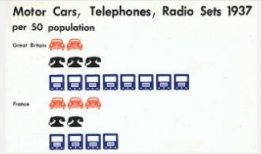
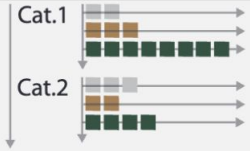
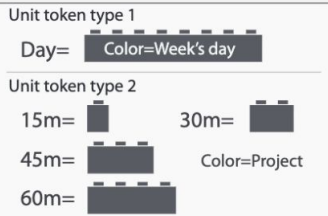
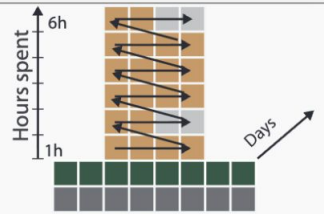
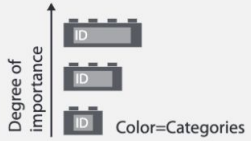
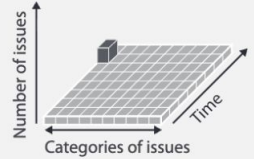
Samuel Huron, Yvonne Jansen, Sheelagh Carpendale



Fig. 1. Constructing a visualization with tokens: right hand positions tokens, left hand points to the corresponding data.

Huron, Samuel, Yvonne Jansen, and Sheelagh Carpendale. "Constructing visual representations: Investigating the use of tangible tokens." IEEE transactions on visualization and computer graphics 20, no. 12 (2014): 2102-2111.

Research about data physicalization

Examples	Picture fragment	Type	C1: Token	C2: Token Grammar	C4: Assembly model	C3: Environment
1. Chris Jordan		Artistic	Object / picture of object	1 picture: = 1 plastic cup, used on airline flights in the US during last six hours	Artistic, the assembly model in this case does not follow the definition of a monosemic system. The assembly is not described as processing the data, but as providing a feeling about the data.	2D Paper canvas
2. Otto Neurath		Analytic	Pictogram	3 Pictograms: = 1 car per 50 people, = 1 bus per 50 people, = 1 phone per 50 people		2D Paper canvas
3. Michael Hunger		Analytic	Lego bricks			3D Physical tangible Lego board
4. Kevin Quinn		Analytic	Lego bricks			3d Physical tangible Lego board

Huron, Samuel, Sheelagh Carpendale, Alice Thudt, Anthony Tang, and Michael Mauerer. "Constructive visualization." In Proceedings of the 2014 conference on Designing interactive systems, pp. 433-442. 2014.

Paradigms of authoring visualization

Visual mapping paradigm	Keep it Simple	Enabling expressivity	Incorporating data dynamics	Manipulation of Visual	Skills learning	CS System examples	NofP
Using	✓	✗	✓	Indirect	Medium	Excel, Spotfire [5], Tableau [6], Google Chart [2] ...	36
Drawing	✓	✓	✗	Direct	Easy/Med	Pen and paper / Illustrator, Photo-shop, ...	13
Coding	✗	✓	✓	Indirect	Hard	Processing, Infovis toolkit [24], Prefuse [31], D3.js [14]...	17
Constructing	✓	✓	✓	Direct	Easy	Unknown	0

Huron, Samuel, Sheelagh Carpendale, Alice Thudt, Anthony Tang, and Michael Mauerer. "Constructive visualization." In Proceedings of the 2014 conference on Designing interactive systems, pp. 433-442. 2014.

Data visualization is powerful to understand data, but also to convey your findings and messages to other audiences.

Try pay attention to the visualizations showing up in your life this week, and see if you can gain inspiration or find mistakes in them.